Newton's Laws: Measured with a Sensor Cart

U3 / S3.1

PREP — BEFORE DAY 1 AND DAY 2

WHEN	TASK	DETAIL
Days before	Charge and test	Charge the 5 sensor carts in use plus the 3 spares, and the 5 separate Go Direct force sensors. Connect each cart to a student device once. Number carts and sensors 1–5 so groups don't grab a neighbour's cart in the connection list.
Days before	Check the signs	On one cart, pull the hook and push the cart forward. Note which sign the force reading gives for a pull and which sign the acceleration gives. If they disagree, Part B slopes come out negative. Decide in advance how you will flip one sign. **Check this in the software yourself; the handout does not name a menu path.**
Before Day 1	Tracks and trays	5 tracks flat on benches, an end stop on each end. Tray per group: cart, one mass bar, 40 cm of string. Balances where every group can reach one.
Before Day 2	Second sensor	Add one force sensor per group with its hook and bumper, plus 10 cm of string. Run the Part C test tug once on your own pair, so you know which sensor needs its sign reversed and how.

Materials — total for 5 groups

Go Direct Sensor Cart × 5 in use (8 working, 3 spare) · Dynamics Track × 5 · end stops × 10 · cart mass bar × 5 · Go Direct force sensor × 5 (Day 2) · string, 40 cm × 5 and 10 cm × 5 · balance, at least 1 · student devices with Graphical Analysis × 5 · track pulley and hanging masses for the extension, as many as you have

TIMING — TWO 50-MINUTE PERIODS

MIN	PHASE	WATCH FOR
D1 0–5	Pre-lab check, trays out, connect carts	Groups connected to the wrong cart number.
D1 5–25	Part A: level, 3 push trials, 3 tow trials	Fit region that includes the push spike. Sensor not zeroed before the tow.
D1 25–45	Part B: cart alone, then cart + bar	Negative slope, which means a sign mismatch. Yanks that hit the end stop.
D1 45–50	Cleanup	Strings off, carts off, mass bars back.
D2 0–5	Connect both sensors, zero both, test tug	Both sensors reading the same sign. Fix it before any data.
D2 5–25	Part C: trials C1–C3 and the C1 stretch	Fingers on the bumper face during C3.
D2 25–45	Post-lab analysis in notebooks	Signs and units on every answer.
D2 45–50	Cleanup	

All five groups run Part C at once with their own cart and force sensor. There is no rotation and no waiting on Day 2.

FAILURE POINTS

- **Force sensor not zeroed.** Zero with nothing on the hook before every run. A leftover offset shifts the Part B intercept and wrecks the Part A tow force, which is only a few hundredths of a newton.
- **Track not level.** A tilt adds a slope component of gravity that swamps friction in Part A. Check that a cart at rest stays put before Part A starts.
- **Sign convention.** If the cart software reports a pull and a forward acceleration with opposite signs, the F–a slope is negative. In Part C, two identical sensors hooked face to face read the *same* sign until one is reversed. Reversing one puts both in track coordinates, and then the minus sign is the "opposite."

- **Fit region.** In Part A students fit across the push and the catch, and the slope is meaningless. Only the coasting stretch counts.
- **String at an angle.** A string pulled up and away puts part of the pull vertical. Keep it level with the track.
- **Force sensor range.** Some Vernier force sensors have a ±10 N / ±50 N range switch. **Unconfirmed for your Go Direct force sensors. Check yours.** If there is one, ±10 N is the better setting here, since nothing in this lab is near 10 N.

Alternatives and backups

Two sensor carts colliding. With 8 working carts and 5 in use, the 3 spares make one extra pair. Two carts meeting bumper to bumper on one track give the same mirror-image force graphs, and both carts have accelerometers, so $m_1a_1 \approx m_2a_2$ can be checked. Good as a demo or for a group that finishes early. **A separate force sensor standing in for a cart.** A force sensor mounted on a plain cart can stand in for one sensor cart's force reading, but that cart has no accelerometer, so the m·a check only works for the sensor cart.

EXPECTED RESULTS — SAMPLE DATA

These numbers are **plausible sample data, not real measurements**. Replace the cart mass with your own balance reading. A good group gets F–a slopes within about 5% of the balance mass and peak-force differences of a few percent. A sloppy-but-passing group gets slopes within about 15% and still sees mirror-image graphs.

QUANTITY	ACCEPTED	SAMPLE	TYPICAL RANGE
Coasting a, level track	0 if no friction	−0.096 ± 0.008 m/s²	small, negative
Tow force at constant v	none	0.034 ± 0.004 N	a few hundredths of a newton
F–a slope, cart alone	0.305 kg (sample balance)	0.318 kg · 4.3%	within 15%
F–a slope, cart + bar	0.802 kg (sample balance)	0.781 kg · 2.6%	within 15%
Peak-force percent difference	0%	2.8% · 3.0% · 3.3%	under about 10%

SAFETY + DISPOSAL

- No chemicals and no glassware are used, so no SDS applies. The safety bullets cover mechanical hazards only: carts rolling off the track end, pinched fingers at the bumper, and a falling hanging mass. **Not yet checked against district policy. Matt to confirm before copying.**
- **Alconox block: kept verbatim, items 4–6, per the locked template rule. It does not fit this lab.** There is no glassware and no goggles are called for, so items 4–6 tell students to do things that don't exist. The S1.3 exception that dropped it is recorded as not a precedent, so dropping it here needs Matt's own decision and its own record. If he says drop it, delete items 4–6 on the handout.

WORKED SAMPLE ANALYSIS — SAMPLE DATA, NOT REAL MEASUREMENTS

Positive is the direction of the push or pull in every part. Sample cart balance mass 0.305 kg. Sample mass bar 0.497 kg, so cart + bar is 0.802 kg.

Part A — first law

Push-trial slopes: −0.089, −0.104, −0.096 m/s². Tow forces: 0.031, 0.038, 0.033 N. Tow-stretch accelerations: +0.004, −0.006, +0.002 m/s².

mean $a = \frac{-0.089 + (-0.104) + (-0.096)}{3} = \frac{-0.289\ \text{m/s}^2}{3} = -0.0963\ \text{m/s}^2$

uncertainty $= \frac{-0.089 - (-0.104)}{2} = \frac{0.015\ \text{m/s}^2}{2} = 0.0075\ \text{m/s}^2$

$a = -0.096 \pm 0.008\ \text{m/s}^2$

mean tow force $= \frac{0.031 + 0.038 + 0.033}{3} = \frac{0.102\ \text{N}}{3} = 0.034\ \text{N}$; uncertainty $= \frac{0.038 - 0.031}{2} = 0.0035\ \text{N}$

$F_{tow} = 0.034 \pm 0.004$ N, with $a = 0.000 \pm 0.005$ m/s² during the tow

Part B — second law

Cart alone: slope 0.318 kg, intercept +0.041 N. percent error $= \frac{|0.318\ \text{kg} - 0.305\ \text{kg}|}{0.305\ \text{kg}} \times 100 = \frac{0.013\ \text{kg}}{0.305\ \text{kg}} \times 100 =$ **4.3%**

Cart + bar: slope 0.781 kg, intercept +0.046 N. percent error $= \frac{|0.781\ \text{kg} - 0.802\ \text{kg}|}{0.802\ \text{kg}} \times 100 = \frac{0.021\ \text{kg}}{0.802\ \text{kg}} \times 100 =$ **2.6%**

Slope rise = 0.781 kg − 0.318 kg = 0.463 kg, against 0.497 kg added: $\frac{|0.463 - 0.497|}{0.497} \times 100 = 6.8\%$ low

Part C — third law

C1 pull: +1.84 N on cart, −1.79 N on hand sensor. Use the sizes. average $= \frac{1.84\ \text{N} + 1.79\ \text{N}}{2} = 1.815\ \text{N}$

percent difference $= \frac{|1.84\ \text{N} - 1.79\ \text{N}|}{1.815\ \text{N}} \times 100 = \frac{0.05\ \text{N}}{1.815\ \text{N}} \times 100 =$ **2.8%**

C2 tugs: +2.31 N and −2.38 N. $\frac{0.07\ \text{N}}{2.345\ \text{N}} \times 100 =$ **3.0%** C3 push: −1.47 N and +1.52 N. $\frac{0.05\ \text{N}}{1.495\ \text{N}} \times 100 =$ **3.3%**

C1 stretch: mean F on cart +0.52 N, mean a +1.58 m/s².

$m \cdot a = (0.305\ \text{kg})(1.58\ \text{m/s}^2) = 0.482\ \text{N}$ $\sum F = F - F_{tow} = 0.52\ \text{N} - 0.034\ \text{N} = 0.486\ \text{N}$

average $= \frac{0.482\ \text{N} + 0.486\ \text{N}}{2} = 0.484\ \text{N}$ percent difference $= \frac{|0.482\ \text{N} - 0.486\ \text{N}|}{0.484\ \text{N}} \times 100 =$ **0.8%**

PRE-LAB KEY

1. Net force on a cart at a constant 0.40 m/s. $\sum F = 0$. Constant velocity means a = 0, and the first law says that only happens when the net force is zero. The push stopped acting the moment the hand left. No credit for "a small forward force keeps it going."
2. Slope of F against a. Mass. Units of slope: $\frac{\text{N}}{\text{m/s}^2} = \frac{\text{kg·m/s}^2}{\text{m/s}^2} = \text{kg}$.
3. Force of cart on sensor when the sensor pulls with +2.0 N. −2.0 N, the same size, pointing back toward the cart. The two forces act on different objects. Only the +2.0 N acts on the cart, so the cart accelerates. Partial credit for −2.0 N with no reason.
4. Part A, step 3. Zero it, with nothing touching the hook. For the tow, the string hangs slack.
5. Region of v–t to fit. Only the coasting stretch after the hand leaves the cart. Leave out the push and the catch.

6. Accepted value in Part B. The balance mass. With a bar on, mass the cart and bar together on the balance. Do not add a stamped value.
7. Part C, step 2. Cart sensor positive, hand sensor negative. If both show the same sign, stop and get the teacher.
8. Safety. End stops on both ends of the track. Hold each sensor by its body, with fingers off the front face and out of the gap.
9. Data tables drawn. Full credit = 3 tables drawn in ink, with units in the row labels.

POST-LAB KEY — FROM THE SAMPLE DATA

1. Coasting a with uncertainty. -0.096 ± 0.008 m/s². Tiny next to the 1–2 m/s² of Part B, so close to zero, but the range −0.104 to −0.088 does not include zero. Rolling friction points opposite the motion, which is the negative direction. "Human error" earns nothing.
2. m|a| against tow force. (0.305 kg)(0.096 m/s²) = 0.029 N, against 0.034 N. While coasting, friction is the only horizontal force, so friction = m|a| (second law). While towing at constant velocity a = 0, so $\sum F = 0$ and the tow force equals friction (first law). Both measure the same friction. Agreement within about 0.005 N is good for forces this small.
3. Percent error for each slope. 4.3% and 2.6%. Full work under Part B of the worked sample analysis, page 3. Units carried in every step.
4. Slope rise and intercept. Up 0.463 kg against 0.497 kg added, 6.8% low. The intercept (+0.041 N, +0.046 N) is the force reading at a = 0, which is the force that only balances friction. It matches the Part A tow force of about 0.03 N. The mass changed, so the slope changed.
5. Percent difference of peaks; signs. 2.8%, 3.0%, 3.3%. The opposite signs mean opposite directions. The forces don't cancel because one acts on the cart and the other acts on the hand sensor. Forces cancel only when they act on the same object.
6. m·a against $F - F_{tow}$. 0.482 N against 0.486 N, 0.8% apart. The force sensor reads only the pull. Friction also acts on the cart, opposite the motion, so $\sum F$ = pull − friction. Common wrong answer: comparing m·a with 0.52 N directly. Partial credit if the friction reason is given.
7. Unzeroed sensor reading +0.15 N. Every force reads 0.15 N high, so the intercept rises by 0.15 N and the slope does not change. Their mass and percent error are unaffected. Common wrong answer: "the mass comes out too big." No credit without the slope-against-intercept reason.
8. Track tilted downhill. Gravity now has a component along the motion. The slope becomes less negative, zero, or even positive, and it hides friction. Part A step 1, the level check with the cart at rest.
9. Connection to 1.3. Part A step 4: the linear-fit slope of v–t is the acceleration. The area under v–t is displacement, in metres.

GRADING

No point split for Physics labs is set yet. Physics gradebook weights are still an open question, so Matt sets this. The carbonless copy is graded individually.

MISSED LAB

If the setup is already torn down, the student completes a Missed Lab Analysis rather than re-running the lab. Give them one group's Tables 1–3 from the same period, plus a printout or screenshot of that group's F–a graph and one Part C force overlay. They answer post-lab questions 1–8 from that data and question 9 from their 1.3 notes.